

Part F – Calculate the Optimum Predictor in MATLAB

After calculating a_{opt} and finding MSE we get these values:

```
Command Window

MSE(a_opt) = 0.096213

MSE(1) = 0.097407

Difference = 0.001195
>>
```

So a_{opt} has a 1.22% improvement over $a = 1$.

Question:

Calculate the prediction error using a_{opt} and compare its MSE with the MSE for $a = 1$. Is the difference significant? Explain based on the characteristics of the signal.

Answer:

No, the difference is not significant. This is because the signal changes gradually and has strong correlation between adjacent samples. Therefore, using $x[n-1]$ with $a = 1$ already provides a good prediction, so optimizing the coefficient only produces a small improvement in the MSE.